Luis Victoria

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Work Experience

Software Engineer — Market Data Platform, Capacity & Performance Feb 2023 – Jul 2026
Bloomberg LP *New York, NY*

Engineered data, performance, capacity, scalability, and reliability systems across Bloomberg's Market Data Platform, spanning Feeds and the Ticker Plant. The platform handled >700B ticks/day during extreme volatility while maintaining <6 ms median and <250 ms p99.9 latency SLAs.

- **Performance Regression Detection**: Applied Bayesian Online Changepoint Detection within Argo Workflows to statistically flag anomalous processing latencies across 30,000+ tick processor instances, cutting remediation time from days to <24 hours.
- **Latency Instrumentation Overhaul**: Authored a C++20 envelope-based timestamping framework tracing tick updates stage-by-stage through the Ticker Plant pipeline with µs resolution, enabling latency regression detection and optimization with <10 µs overhead.
- **Predictive Capacity Forecasting**: Rewrote a fragile legacy log-scraping storage monitor as a two-part Python system: a fleet-wide scanner writing structured per-database storage and machine metadata to a centralized datastore, and a forecasting service extrapolating rolling usage trends 14 days ahead with automated alerting. Delivered in one week during the April 2025 tariff-driven volatility spike, protecting 1,000+ machines from storage-exhaustion/data-loss risk.
- **Production & Incident Ownership**: On-call engineer in 24/7 follow-the-sun rotation for Bloomberg's Market Data Platform; isolated bad data, performed live synchronization and recovery, and drove root-cause analysis for major outages under volatile market conditions.
- **Capacity Planning & Cluster Rollouts**: Assessed CPU, memory, and storage capacity ahead of onboarding new global exchanges; owned staged rollouts of cluster splits and new clusters across regions, offloading overloaded machines and avoiding SLO breaches with minimal client disruption.
- **Disaster Recovery Performance Resilience**: Automated load reassignment during data-center failover; benchmarked CPU thread-pinning under combined load, informing Linux cgroups-based CPU prioritization to preserve broadcast continuity while controlling dropped requests.

Quantitative Developer Intern May 2019 – Aug 2019
Farringdon Asset Management *Singapore*

- **Automation with Bloomberg API**: Reduced time to generate client portfolio analysis by 50% for the portfolio management team using information from PORT<GO>.
- **Backtesting Enhancements**: Extended the firm's Python backtesting engine to support cryptocurrency analysis, integrating market-data pipelines through CCXT and Binance.

Education

Business and Computer Science Sep 2018 – Dec 2022
Bachelor of Commerce, Sauder School of Business, University of British Columbia *Vancouver, BC*

- **Relevant Coursework**: Machine Learning & Data Mining, Matrix Algebra, Calculus III, Empirical Economics, Advanced Algorithmic Design, Advanced Relational Databases.
- **Sauder Trading Simulation**: Built a Raspberry Pi arbitrage bot for a non-matching order-book simulation, polling every 2 min and executing crossed bid/ask opportunities; placed 1st of 136 students.

Skills

Programming Languages: C++20, Python, SQL (PostgreSQL)
Tools: Linux (cgroups), Grafana, Humio (LogScale), Jenkins, Git, pytest, pydantic, uv, Ruff, Bloomberg Terminal